

Lecture 8 - PID Control

Instructor: Martin Nisser

Scribe: Catherine Ingram (Please send feedback to ingramca@uw.edu)

8.1 PID Control

Popular control design combining Proportional, Integral, and Derivative terms. The control input u is a weighted sum of the error, its integral, and its derivative.

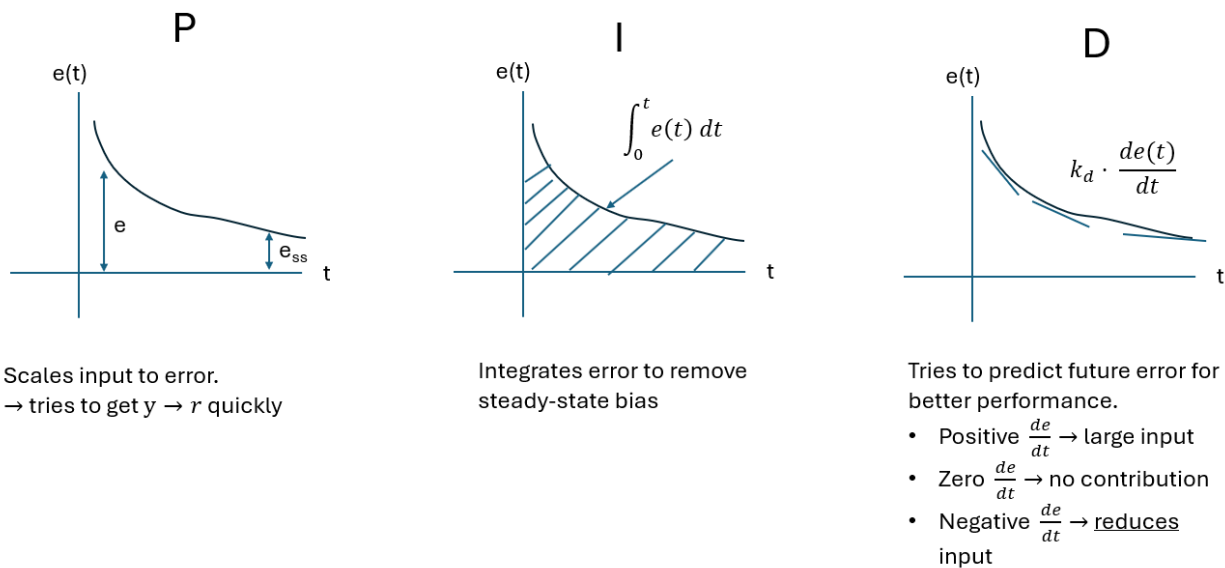


Figure 8.1: Breakdown of each term in PID control.

$$u(t) = k_p e(t) + k_I \int_0^t e(t) dt + k_D \cdot \frac{de(t)}{dt}$$
$$U(s) = E(s) \cdot \left(k_p + \frac{k_I}{s} + k_D s \right)$$

- Derivative acts like a "brake", improving the transient response (if properly tuned, by reducing overshoot & settling time).
- Derivative gives sharp response to Δ error $\rightarrow$ can amplify noise (D is avoided if high-frequency noise is expected).
- Derivative control is never used on its own.

Q: Can you guess why?

Ans: Derivative contains no information on the value of the error! $\rightarrow$ If $e(t)$ is large, but not changing, no control input would occur.

$\rightarrow$ Typically, we do $\underbrace{P}_{\text{basic control}} \rightarrow \underbrace{PI}_{\text{Remove } E_{ss}} \rightarrow \underbrace{PID}_{\text{Improve performance}}$

Implications of PID

Let's consider adding a PID controller to 1st order plant $G = \frac{1}{s+1}$. We'll look at:

1. Our ability to shape the transfer function G_{RY} by placing its poles (i.e. setting ω_n and ζ).
2. How we affect the system type number.

1. Pole Placement

Lets find GH first:

$$\begin{aligned} H(s) &= k_p + \frac{k_I}{s} + k_D s, \quad G(s) = \frac{1}{s+1} \\ GH &= (k_p + \frac{k_I}{s} + k_D s) \cdot \frac{1}{s+1} \\ &= \frac{sk_p + k_I + k_D s^2}{s} \cdot \frac{1}{s+1} \\ &= \frac{k_D s^2 + k_p s + k_I}{s(s+1)} \end{aligned}$$

Now, plug into G_{RY} :

$$\begin{aligned} G_{RY} &= \frac{GH}{1+GH} \\ &= \frac{\frac{k_D s^2 + k_p s + k_I}{s(s+1)}}{1 + \frac{k_D s^2 + k_p s + k_I}{s(s+1)}} \\ &= \frac{k_D s^2 + k_p s + k_I}{s^2 + s + k_D s^2 + k_p s + k_I} \\ &= \frac{k_D s^2 + k_p s + k_I}{(k_D + 1)s^2 + (k_p + 1)s + k_I} \\ &= \left(\frac{1}{k_D + 1} \right) \frac{k_D s^2 + k_p s + k_I}{s^2 + \left(\frac{k_p + 1}{k_D + 1} \right) s + \left(\frac{k_I}{k_D + 1} \right)} \end{aligned}$$

Q: Is PID necessary? Or might PI have been sufficient?

Ans: There are only 2 poles, but we have 3 DoF, i.e. 3 control variables to choose. We only need two (P & I) to place the poles or set our ω_n and ζ .

$\therefore$ PID is probably overkill for 1st order G, but will let us place all 3 poles for 2nd order G.

2. Type Number

We've seen that system type is defined by the number of $\frac{1}{s}$ in GH, and dictates the degree of the polynomial it can track.

Recall:

$$r(t) = \frac{t^k}{k!} \rightarrow e(t)_{t \rightarrow \infty} = \lim_{s \rightarrow 0} \frac{1}{s^k} S(s)$$

In our system:

$$\begin{aligned} S &= \frac{1}{1 + GH} \\ S &= \frac{1}{1 + \frac{k_D s^2 + k_p s + k_I}{s(s+1)}} \\ &= \frac{s(s+1)}{(k_D + 1)s^2 + (k_p + 1)s + k_I} \end{aligned}$$

Note: System type when using PID is +1 larger than G, because of the common integrator in: $\frac{sk_p + k_I + s^2 k_D}{s}$.

→ System Type = 1 (same as PI controller).

→ Both PID & PI make this system Type 1.

→ P control would leave this system as Type 0.

8.2 Examples: Apply $P \rightarrow PI \rightarrow PID$ Control

8.2.1 Position Control of a DC Motor

$$\begin{aligned} \underbrace{T}_{\text{Torque}} - \underbrace{b}_{\text{friction}} \dot{\theta} &= \underbrace{I}_{\text{Inertia}} \ddot{\theta} \\ I \ddot{\theta} + b \dot{\theta} &= T \\ G(s) = \frac{\Theta(s)}{T(s)} &= \frac{1}{Is^2 + bs} = \frac{1}{s(Is + b)} = \frac{1/I}{s(1s + \frac{b}{I})} = \underbrace{\frac{A}{s(s + \tau)}}_{A, \tau > 0} \\ A = \frac{1}{I}, \tau &= \frac{b}{I} \end{aligned}$$

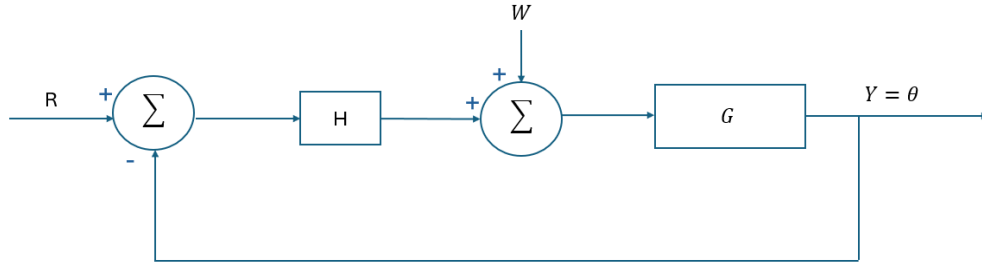


Figure 8.2: DC motor system block diagram with P control and external disturbance W.

1. $H(s) = k_p$: P control

$$G_{RY} = \frac{GH}{1+GH} = \frac{\frac{k_p A}{s(s+\tau)}}{1 + \frac{k_p A}{s(s+\tau)}} = \frac{k_p A}{s^2 + \tau s + \underbrace{k_p A}}$$

Stability? $\rightarrow k_p > 0$ for stability

Pole placement? $\rightarrow$ can set $\omega_n^2 = \frac{k_p A}{\tau}$, but cannot set ω_n and ζ independently

System Type

- Sys type of $G_{RE}(s) = S(s) \leftarrow$ Tracking (G_{RE} is the T.F. from reference to error)

$$G_{RE} = S(s) = \frac{1}{1+GH} = \frac{1}{1 + \frac{k_p A}{s(s+\tau)}} = \frac{s(s+\tau)}{s^2 + \tau s + k_p A}$$

$\rightarrow$ Sys type = 1 $\leftarrow$ Can track: step with 0 error, ramp with bounded error

- Sys type of $G_{WE}(s) \leftarrow$ Disturbance rejection

$$G_{WE} = \frac{G}{1+GH} = \frac{\frac{A}{s(s+\tau)}}{1 + \frac{k_p A}{s(s+\tau)}} = \frac{A}{s^2 + \tau s + k_p A}$$

$\rightarrow$ Sys type = 0 $\leftarrow$ Can reject constant disturbance with bounded error

(Attenuated by: $\frac{A}{k_p A}$), Cannot reject ramp disturbance

2. $H(s) = k_p + \frac{k_I}{s}$: PI control

Let's focus on characteristic equation $\leftarrow$ common denominator to G_{RY}, G_{RE}, G_{WE}

$$1 + GH = 1 + \underbrace{\frac{k_p s + k_I}{s}}_H \cdot \underbrace{\frac{A}{s(s+\tau)}}_G \left\{ \begin{array}{l} \text{Q: Can we already see sys type for } G_{RE}, G_{WE}? \\ G_{RE} = \frac{1}{1+GH} \rightarrow \text{get } s^2 \text{ in numerator: sys type} = 2 \\ G_{WE} = \frac{G}{1+GH} \rightarrow \text{get } s \text{ in numerator: sys type} = 1 \end{array} \right.$$

$$= s^3 + \tau s^2 + A k_p s + A k_I$$

For stability: $k_p > 0$, $k_I > 0$ AND $\tau A k_p > A k_I \rightarrow \tau k_p > k_I$

For Pole placement: Cannot set location of all 3 poles with just 2 DoF (k_p & k_I)

3. $H(s) = k_p + \frac{k_I}{s} + k_D s$: PID

$$1 + GH = 1 + \frac{k_p s + k_I + k_D s^2}{s} \cdot \frac{A}{s(s+\tau)} \quad \left. \vphantom{\frac{k_p s + k_I + k_D s^2}{s}} \right\} \text{ Sys type same as for PI}$$

$$= s^3 + \frac{(k_D A + \tau) s^2}{\tau} + A k_p s + A k_I$$

$\rightarrow$ For stability: $k_D > \frac{-\tau}{A}$, $k_p > 0$, $k_I > 0$ AND $(k_D A + \tau) k_p A > k_I A \Rightarrow (k_D A + \tau) k_p > k_I \leftarrow k_I$ cannot be very big

$\rightarrow$ Pole placement: We have 3 poles, and 3 DoF: k_p , k_D , k_I

$\therefore$ Have ability to place all 3 poles